Software Test Plan (STP) - Pocket Packet Sniffer

Project: Pocket Packet Sniffer — Wi-Fi / Ethernet Packet Sniffing & Analysis Dashboard
Version: 1.0
Authors: CHINMAYI HEBBAR AM, CHENGAPA KAIBILIRA SANTOSH, GOWTHAM L, HAMZA SHABBIR SAHAPURWALA
Date: 03-09-2026
Status: Draft v1.0

1. Introduction

Purpose: This document defines the test plan for the Pocket Packet Sniffer (PPS) v1.0. It outlines test objectives, scope, strategy, environment, schedule, responsibilities, risks, and traceability for verification of the software requirements.

Scope: Testing covers packet capture from available Wi-Fi and Ethernet interfaces; interface selection and capture filtering; packet processing and Zeek-based traffic analysis; anomaly detection; live dashboard visualization and session controls; local storage, retention, session history, and export; system management; authentication; performance, portability, reliability, usability, maintainability, scalability/storage; and the defined security requirements.

Out of scope: cloud-hosted multi-tenant deployment, hardware integration, decryption of TLS-protected payloads, and long-term (>90 day) data warehousing.

References: Pocket Packet Sniffer Software Requirements Specification (SRS) v1.0, dated 03-09-2026; Section 4 System Features; Section 5 Non-Functional Requirements and Security; Section 6 Quality Attributes & Acceptance Tests; Section 8 Requirements Traceability Matrix.

Definitions: PPS (Pocket Packet Sniffer), SRS (Software Requirements Specification), RTM (Requirements Traceability Matrix), Wi-Fi (IEEE 802.11 wireless local area networking), Zeek (network security monitoring / traffic analysis framework), PCAP (packet capture file format), BPF (Berkeley Packet Filter), UI (User Interface), API (Application Programming Interface), TLS (Transport Layer Security).

2. Test Items

- Packet Capture module (Wi-Fi/Ethernet capture, interface selection, capture filters)
- Traffic Analysis & Zeek Integration module
- Packet normalization and host/device profiling
- Web Dashboard & Visualization
- Data Storage, Retention, Export, and Session History
- System Management and User Operations
- Access Control / Authentication
- Security controls and audit logging
- Backend REST/WebSocket interfaces
- Local SQLite/flat-file data store

3. Features to be Tested

Features mapped to SRS requirement IDs and the test-case references defined in the individual requirement tables:

Packet Capture
- PPS-F-001 → TC-CAP-01 — Wi-Fi packet capture on the selected network interface
- PPS-F-002 → TC-CAP-02 — Start and stop packet capture sessions
- PPS-F-003 → TC-CAP-03 — Ethernet traffic capture through an inline tap or mirrored switch port when connected via RJ45
- PPS-F-004 → TC-CAP-04 — Select and switch between Wi-Fi/Ethernet capture interfaces
- PPS-F-005 → TC-CAP-05 — BPF-style capture filters by protocol, host, or port

Traffic Analysis & Zeek Integration
- PPS-F-006 → TC-ZEEK-01 — Feed captured PCAP data to Zeek for near-real-time analysis
- PPS-F-007 → TC-ZEEK-02 — Generate applicable standard Zeek logs for captured traffic
- PPS-F-008 → TC-ZEEK-03 — Detect and flag anomalous or policy-defined suspicious traffic patterns
- PPS-F-009 → TC-ZEEK-04 — Normalize Zeek logs and packet metadata for dashboard/reporting use
- PPS-F-010 → TC-ZEEK-05 — Identify and display active host information and traffic statistics

Dashboard & Visualization
- PPS-F-011 → TC-DASH-01 — Live traffic statistics
- PPS-F-012 → TC-DASH-02 — Active network hosts and captured packet information
- PPS-F-013 → TC-DASH-03 — Start/stop/pause capture sessions and select network interfaces through UI controls
- PPS-F-014 → TC-DASH-04 — Display Zeek-flagged security alerts and anomalies

Data Storage & Export
- PPS-F-015 → TC-DATA-01 — Local storage of PCAP files and Zeek logs with configurable retention
- PPS-F-016 → TC-DATA-02 — Export PCAP files, Zeek logs, and CSV reports
- PPS-F-017 → TC-DATA-03 — Session naming/tagging and historical session browsing

System Management & User Operations
- PPS-F-018 → TC-SYS-01 — Configure and save packet-capture settings and filters
- PPS-F-019 → TC-SYS-02 — Authenticated access to the web dashboard through a supported browser
- PPS-F-020 → TC-SYS-03 — Safe start/stop/termination of packet capture sessions without data loss or corruption

Access Control
- PPS-F-021 → TC-AUTH-01 — Authentication required before dashboard and packet-capture management access

Non-Functional Requirements
- PPS-NF-001 → TC-PERF-01 — Dashboard live-statistics latency ≤ 2 seconds under normal capture load up to 20 Mbps
- PPS-NF-002 → TC-COMP-01 — Windows/Linux compatibility with Python 3.x and Zeek installed
- PPS-NF-003 → TC-REL-01 — At least 4 hours of continuous packet capture without crashes,

memory leaks, or data loss
- PPS-NF-004 → TC-UX-01 — Responsive dashboard on desktop and mobile browsers
- PPS-NF-005 → TC-MAINT-01 — Modular architecture with documented interfaces
- PPS-NF-006 → TC-STOR-01 — Configurable retention/log rotation and capture sessions up to 5 GB without noticeable performance degradation

Security Requirements
- PPS-SR-001 → TC-SEC-01 — Secure password hashing and no plaintext passwords in storage/logs
- PPS-SR-002 → TC-SEC-02 — HTTPS/TLS protection of dashboard-backend communication where supported
- PPS-SR-003 → TC-SEC-03 — Access restriction for stored PCAP files, Zeek logs, and exported reports
- PPS-SR-004 → TC-SEC-04 — Secure initial administrator credentials and no hardcoded credentials/secrets
- PPS-SR-005 → TC-SEC-05 — Redaction/masking of sensitive captured payloads in default dashboard views
- PPS-SR-006 → TC-SEC-06 — Audit logging of administrative actions

4. Features Not to be Tested

- Cloud-hosted multi-tenant deployment
- Hardware integration and dedicated hardware interfaces
- Decryption of TLS-protected payloads
- Long-term (>90 day) data warehousing

5. Test Approach / Strategy

Levels:
- Unit testing (module-level verification of capture, analysis, backend, dashboard, and data-storage logic)
- Integration testing (packet capture → Python processing → Zeek → backend/dashboard/storage interfaces)
- System testing (end-to-end authenticated packet capture, analysis, visualization, storage, and export workflows)
- Acceptance testing against the SRS acceptance criteria and defined test suites

Types:
- Functional testing of PPS-F-001 to 021
- Non-functional testing of PPS-NF-001 to 006
- Security testing of PPS-SR-001 to 006 and PPS-F-021
- Regression testing after defect fixes and integration changes
- Performance and reliability testing under the defined load/duration targets
- Usability and responsive-layout testing on supported desktop/mobile browsers
- Compatibility/portability testing on supported Windows and Linux environments

Entry Criteria: A testable build is available; the required test environment and dependencies are available (including supported network interfaces, Python 3.x, Zeek, browser, and local storage); controlled/lab network traffic and test scenarios are available; and the relevant implementation is ready for execution.

Exit Criteria: All planned test cases have been executed; all High-priority functional requirements are implemented and verified; no critical/High-priority NFR or security-requirement failures remain outstanding; and the RTM shows the listed test cases passed (status = P), consistent with the SRS exit criteria.

5.1 Security Validation

- Validate PPS-SR-001 credential handling, including salted industry-standard password hashing and absence of plaintext passwords in storage/logs.
- Validate PPS-SR-002 HTTPS/TLS protection of dashboard-backend communication where supported.
- Validate PPS-SR-003 access restrictions on stored PCAP files, Zeek logs, and exported reports.
- Validate PPS-SR-004 secure first-run administrator credential creation and absence of hardcoded credentials/secrets.
- Validate PPS-SR-005 masking/redaction of sensitive captured payloads in default dashboard views.
- Validate PPS-SR-006 audit logging for session start/stop, configuration changes, exports, and login attempts.
- Validate PPS-F-021 protected dashboard and packet-capture management access for unauthenticated users.

6. Test Environment

Hardware: Host computer with available Wi-Fi and/or Ethernet network interfaces; Ethernet test access may use an inline tap or mirrored switch port as specified by PPS-F-003. No dedicated hardware interface is required by the SRS.
Software: Windows/Linux, Python 3.x, Zeek, SQLite and/or flat-file local storage, Flask/FastAPI backend, modern web browser, and the PPS application.
Interfaces/Communications: RESTful APIs for dashboard requests and system management; WebSocket updates for live packets, traffic statistics, and security alerts; HTTP/HTTPS on the local host or LAN.
Tools/Utilities: Controlled lab traffic/test devices, packet-capture tooling required by the implementation, browser, Zeek, and the project's Jira/GitHub workflow where applicable. The SRS does not prescribe specific UI automation or performance-tool products.
Test Data: Controlled network traffic covering Wi-Fi/Ethernet packets, relevant protocol traffic for Zeek logs, simulated suspicious traffic such as a port scan, sensitive plaintext-credential test traffic for masking validation, session metadata, and administrative actions.

7. Test Schedule

Milestones aligned to the SRS sprint plan:
- Sprint 1: Development environment setup, Python packet-capture setup, Zeek installation/configuration, project skeleton, Jira board and GitHub repository configuration.
- Sprint 2: Core Wi-Fi/Ethernet packet capture working end-to-end into PCAP; initial dashboard wireframe/mock data; authentication scaffolding.
- Sprint 3: Zeek analysis pipeline with real parsed data; protocol analysis, alerts, active-host information, and integrated security requirements.
- Sprint 4: Performance and reliability testing, RTM sign-off, bug fixing, end-to-end integration, final documentation, and demo.

Specific calendar execution dates are not defined in the SRS and are therefore not assumed here.

8. Test Deliverables

- Software Test Plan (this document)
- Functional, non-functional, and security test cases
- Test scripts / execution procedures as applicable
- Controlled test data and lab traffic scenarios
- Test execution logs
- Defect reports
- Requirements Traceability Matrix updates / verification evidence
- Test Summary Report

9. Roles and Responsibilities

Role	Name	Responsibility
Team Lead / System Architect	CHINMAYI HEBBAR AM	System integration, architecture, Jira & GitHub administration, security; owns PPS-F-021, PPS-SR-001 to 006, PPS-NF-005
Embedded & Network Capture Engineer	CHENGAPA KAIBILIRA SANTOSH	Python-based packet capture, Wi-Fi/Ethernet interface configuration, packet filtering, packet logging, backend integration; owns PPS-F-001 to 005, PPS-F-018 to 020, PPS-NF-002
Packet Analysis Engineer	GOWTHAM L	Python packet processing, Zeek integration, protocol analysis, threat detection, alerts, traffic statistics; owns PPS-F-006 to 010, PPS-F-015 to 017, PPS-NF-006
Dashboard Engineer / QA	HAMZA SHABBIR SAHAPURWALA	Dashboard development, data visualization, frontend integration, testing, RTM, documentation; owns PPS-F-011 to 014, PPS-NF-001, PPS-NF-004

10. Risks and Mitigation

Risk	Mitigation
Zeek unavailable or misconfigured	Verify/install/configure Zeek before Zeek-dependent tests; execute Zeek test cases only in a valid environment.
Required Wi-Fi/Ethernet interface or capture permissions unavailable	Use a supported accessible interface and ensure appropriate permissions before packet-capture execution.

Resource limits affect sustained Zeek analysis throughput	Include performance, reliability, and storage tests against the SRS targets; monitor failures and data integrity.

11. Assumptions & Dependencies

- Zeek must be installed and operational for Zeek-related tests.
- Valid and accessible network interfaces and appropriate packet-capture permissions are required.
- Testing must use networks/devices owned by the operator or for which explicit permission exists; demonstrations should use controlled lab traffic and test devices.
- The supported software environment is Windows/Linux with Python 3.x, Zeek, SQLite/flat files, a Flask/FastAPI backend, and a modern web browser.
- The SRS defines local deployment; cloud-hosted multi-tenant deployment is out of scope.
- Test execution depends on the availability of the required implementation modules and their interfaces.

12. Suspension & Resumption Criteria

Suspend testing if:

- A required test environment component or dependency (for example, the target network interface, Zeek installation, supported runtime, browser, or local storage) is unavailable and prevents valid execution.
- A blocking defect or unstable build prevents meaningful execution of the affected test set.
- Test traffic or devices required for a controlled scenario are unavailable.

Resume testing if:

- The blocking defect is resolved or an executable build is restored.
- The required environment/dependency is available and stabilized.
- The required controlled test scenario is available again.

13. Test Case Management & Traceability

RTM ensures mapping of SRS requirements to test cases. For this Test Plan, the individual requirement-table mappings in SRS Sections 4 and 5 are the authoritative test-case references.

Examples:

- PPS-F-001 → TC-CAP-01
- PPS-F-008 → TC-ZEEK-03
- PPS-F-014 → TC-DASH-04
- PPS-F-016 → TC-DATA-02
- PPS-F-019 → TC-SYS-02
- PPS-F-021 → TC-AUTH-01
- PPS-NF-001 → TC-PERF-01
- PPS-NF-002 → TC-COMP-01
- PPS-SR-001 → TC-SEC-01

Note: The SRS RTM in Section 8 uses alternate labels for PPS-F-018 to 020 and PPS-NF-002 (TC-PWR-01 to 03 and TC-PORT-01). Per the stated project instruction for this Test Plan, the mappings above follow the individual requirement tables in Sections 4–5 instead.

14. Test Metrics & Reporting

Metrics collected:
- % test cases executed
- % test cases passed/failed
- Requirement coverage across PPS-F, PPS-NF, and PPS-SR IDs
- Defect count/severity and defect aging
- Functional acceptance-criteria pass rate
- Performance latency against PPS-NF-001
- Reliability duration/data-integrity results against PPS-NF-003
- Storage/retention results against PPS-NF-006

Reports:
- Test execution status
- Defect reports / triage status
- Requirements traceability status
- Final Test Summary Report

15. Approvals

Role	Name	Signature / Date
Team Lead / System Architect	CHINMAYI HEBBAR AM	14-09-2026
Embedded & Network Capture Engineer	CHENGAPA KAIBILIRA SANTOSH	14-09-2026
Packet Analysis Engineer	GOWTHAM L	14-09-2026
Dashboard Engineer / QA	HAMZA SHABBIR SAHAPURWALA	14-09-2026